

Nicholas Wong

nicwjh@mit.edu — nicwong.com — [LinkedIn](#)

EDUCATION

Massachusetts Institute of Technology	2026 MFin , Master of Finance GPA: 4.6/5.0
UNC Chapel Hill	2024 B.S. , Computer Science; B.S. , Economics; Minor , Statistics and Analytics GPA: 3.95/4.0; Graduated with Highest Distinction (summa cum laude equivalent)
Stanford University	2026–28 Visiting Research Fellow (Non-Degree)
Harvard University	2024 Cross-Registration (Non-Degree) GPA: 4.0/4.0

Selected Coursework: Real Analysis; Probability Theory; Linear Algebra; Differential Equations; Optimization; Advanced Econometrics; Time Series Analysis; Machine Learning; Deep Learning; Advanced Financial Mathematics; Fixed Income & Derivatives; Options & Futures; Data Structures; Algorithms & Analysis

RESEARCH EXPERIENCE

My research interests lie broadly within economics and computation. Currently, my interests include algorithmic economics, empirical finance, and applied AI/ML.

Massachusetts Institute of Technology Sloan School of Management (Cambridge, MA, USA)	2026	Research Assistant Supervisor: Dr. Haoxiang Zhu
Verition Fund Management Fixed Income & Macro Group (New York, NY, USA)	2025	Macro Research Intern Supervisor: Nisarg Kamdar (Portfolio Manager)

A Quantitative Framework for Macroeconomic Forecasting

- Built a real-time forecasting system that predicts surprise of U.S. Nonfarm Payrolls using 20+ years of data with statistical significance. (Python)

- Developed uncertainty bands around each prediction using four statistical models (density estimation), including a novel dynamic multiplier that widens prediction bands when forecaster disagreement signals elevated tail risk. Prediction bands were well-calibrated and matched their target confidence levels within 1–2% on average.

MIT–Panagora Asset Management	2025	Research Practicum
Joint Finance Lab (Cambridge, MA, USA)		Supervisors: Dr. Bhushan Vartak, Dr. Gita Rao

Large Language Models in Equity Research

- Built a rolling-window XGBoost classification framework to predict earnings changes and surprises for Russell 3000 stocks using 153 asset pricing factors, financial statement data, and LLM-generated embeddings (direction, magnitude, confidence, narrative text embeddings from three open-source models). Found that LLM features improved classification accuracy most when paired with financial statement inputs, suggesting LLMs provide complementary narrative context beyond raw numerical data.
- Constructed long-short tercile portfolios from XGBoost prediction signals, demonstrating that LLM-augmented strategies achieved up to 20% annualized alpha with information ratios above 1.0, significantly outperforming both base factor models and consensus analyst forecasts across multiple lag specifications.
- Implemented chain-of-thought prompting pipelines across three open-source LLMs (Llama-3-8B, DeepSeek-R1-Distill variants) to generate earnings predictions and financial narrative analyses from anonymized Compustat balance sheets and income statements, contributing to prompt design, data pipeline construction, and performance evaluation against IBES analyst benchmarks. (Python, XGBoost, HuggingFace)
- Offered quantitative research summer 2025 internship opportunity at PanAgora Asset Management (equity investment team, stock selector equity strategies)

MIT–T. Rowe Price	2024	Research Practicum
Joint Proseminar in Capital Markets (Cambridge, MA, USA)		Supervisors: Mark Kritzman, Sébastien Page (CIO)

Artificial Intelligence for Financial Analysis

- Designed a three-layered empirical framework with escalating prompt complexity to evaluate GPT-4o’s ability to generate sell-side equity research reports for pure-play and multi-segment companies.
- Benchmarked AI-generated reports against human analyst reports using a rubric-based utility evaluation and NLP methods (TF-IDF keyword extraction, cosine similarity), finding that human analysts consistently outperform across depth, accuracy, and actionability. (Python)
- Conducted a literature review synthesizing recent findings on LLM performance in quantitative financial analysis, including chain-of-thought prompting, earnings prediction, and human-AI complementarity.

RESEARCH PROJECTS

- **Double Descent in Financial Time Series.** Nicholas Wong. Individual research project for *6.7960 Deep Learning*, Massachusetts Institute of Technology. Course supervised by Dr. Kaiming He, Dr. Sara

Beery, Dr. Omar Khattab. Empirical study of generalization behavior in overparameterized neural networks applied to synthetic and financial time series data, examining the conditions under which double descent emerges and fails in low signal-to-noise environments.

- **Portfolio Optimization.** Nicholas Wong. Individual research project conducted at Harvard University for *Machine Learning and Big Data*, supervised by Dr. Soroush Saghaian. Developed an ensemble machine-learning-based forecasting and sparsified Markowitz optimization framework for NASDAQ-100 equities, achieving improved risk-adjusted performance relative to market benchmarks.
- **Regulatory Response to the SVB Collapse: A Natural Language Processing Analysis.** Nicholas Wong and Priscilla Clark. Research project conducted at the Massachusetts Institute of Technology; supervised by Dr. Mike Chen and Dr. Andrew Zachary. Applied natural language processing techniques to analyze shifts in U.S. banking regulatory discourse before and after the Silicon Valley Bank collapse, testing research hypotheses and documenting increased emphasis on liquidity, capital adequacy, and systemic risk oversight.

TEACHING EXPERIENCE

Massachusetts Institute of Technology MIT EECS (Cambridge, MA, USA)	2025	Laboratory Assistant Intro. to Programming and Computer Science
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- Held weekly office hours and coached students on Python programming fundamentals, debugging techniques, and best practices; assisted in course delivery by reviewing problem sets and model solutions prior to release.

AWARDS

- Phi Beta Kappa, 2023.
- Dean's List (multiple awards), 2021-2023.
- Merit Scholarship, New York University.

LEADERSHIP

- Former Reconnaissance Team Leader, Singapore Armed Forces; led reconnaissance platoon of 30 soldiers in combat training and exercises.

MISCELLANY

- Technical Skills: Python (Pandas, Polars, NumPy, Scikit-learn, Keras, TensorFlow, PyTorch), R, Stata, SQL, Java; Git; Linux/Unix; LaTeX; Bloomberg Terminal.
- Certifications: Machine Learning Specialization (Stanford University), Data Science Credential in Economics (UNC Chapel Hill), AI Fluency: Frameworks and Foundations (Anthropic), ChatGPT Prompt Engineering for Developers (DeepLearning.AI, OpenAI).

- Standardized Test Scores: GRE Quantitative Reasoning 169/170; SAT 1540/1600
- Activities: former competitive strength athlete (powerlifting) and competitive badminton player.
- Interests: fitness, barbell sports, health & wellness, cooking, history.